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%EA_HW_3_6 Nouh Shaikh
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```
clc; clear; close all;
```

```
x = [3,4,5,7,8,9,11,12];
```

```
y = [1.6, 3.6, 4.4, 3.4, 2.2, 2.8, 3.8, 4.6];
```

```
n = length(x);
```

```
Z = [x.^3; x.^2; x; ones(1, n)]';
```

```
a = (Z' * Z) \ (Z' * y');
```

```
fprintf('Coefficients (a3 to a0): %.4f, %.4f, %.4f, %.4f\n', a);
```

```
y_pred = Z * a;
```

```
Sr = sum((y' - y_pred).^2);
```

```
St = sum((y - mean(y)).^2);
```

```
r2 = 1 - (Sr / St);
```

```
syx = sqrt(Sr / (n - 4));
```

```
fprintf('r^2 = %.4f, s_y/x = %.4f\n', r2, syx);
```

```
x_fit = linspace(min(x), max(x), 100);
```

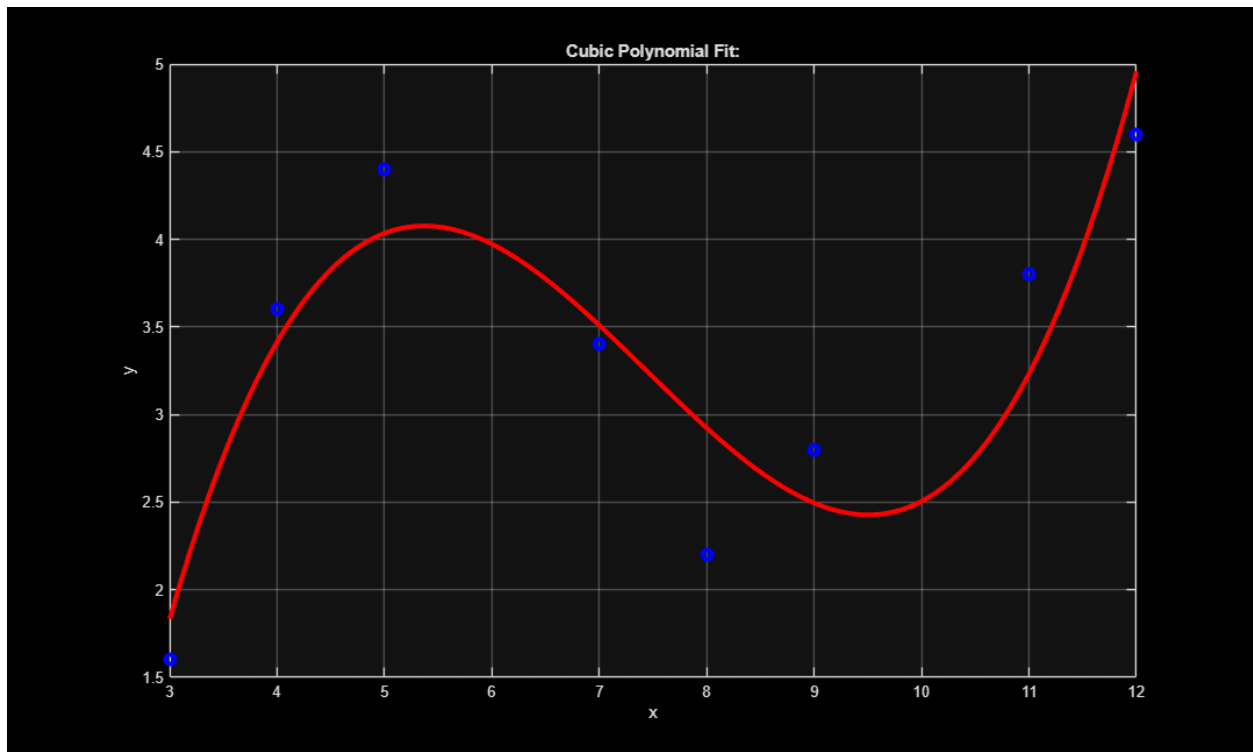
```
y_fit = polyval(a, x_fit);
```

```
plot(x, y, 'bo', x_fit, y_fit, 'r-', 'LineWidth', 3);
```

```
grid on; xlabel('x'); ylabel('y'); title('Cubic Polynomial Fit:');
```

```
Coefficients (a3 to a0): 0.0467, -1.0412, 7.1438, -11.4887
```

```
r^2 = 0.8290, s_y/x = 0.5700
```



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